

Chapter 1

Functions

A relation is a collection of points. A function is a relation in which each input determines a single output, which is precisely the condition that lets you predict. Data arrives as a scatter of points, and the work of the subject is to find a predictable rule running through it: a function fit to the cloud, reshaped from a known curve through $A f(b(x - h)) + k$.

1.1 Relations as Point Clouds

Record, for every student in a class, their height and their shoe size. Each student becomes a pair of numbers: an input (height) and an output (shoe size). Plotted, these pairs form a cloud of dots. The cloud is the raw object of study. It carries no formula and no curve; it is the data, exactly as measured.

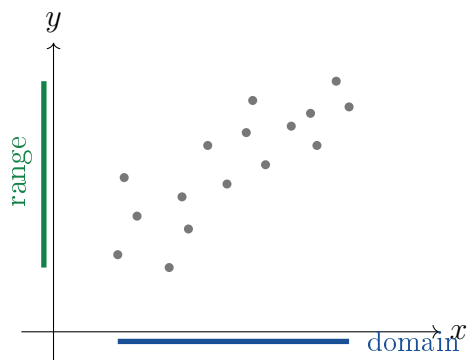
Definition: Relation. A *relation* is any set of ordered pairs (x, y) . The set of all first coordinates is the *domain*; the set of all second coordinates is the *range*.

A relation can be specified by a list, a table, an equation, or a graph. These are four ways of naming the same set of pairs. The list

$$\{(1, 2), (3, 5), (3, -1), (4, 0)\}$$

is a relation with domain $\{1, 3, 4\}$ and range $\{2, 5, -1, 0\}$. So is the set of all (x, y) satisfying $x^2 + y^2 = 1$; that relation is a circle, an infinite set of points packed densely enough to read as a curve.

The picture. A relation is a cloud of dots suspended above the x -axis. The domain is the shadow the cloud casts straight down onto the x -axis; the range is the shadow it casts sideways onto the y -axis. Domain and range are the two shadows of the cloud.



Example 1.1.1. (*reading domain and range from a list*) For $\{(-2, 4), (0, 0), (1, 1), (2, 4)\}$ the domain is $\{-2, 0, 1, 2\}$ and the range is $\{0, 1, 4\}$. The value 4 appears twice in the range but is listed once; a range is a set, so repeats collapse.

1.2 When a Relation Is a Function

Most clouds are unstructured. A few satisfy a strong condition: each input is paired with exactly one output. That condition is what makes prediction possible. Given the input, there is one permitted output, so the output can be stated. If two outputs share an input, prediction fails, because the input no longer selects a single value.

Definition: Function. A *function* is a relation in which each input is paired with exactly one output. Equivalently, no two distinct pairs share the same first coordinate.

The relation $\{(1, 2), (3, 5), (3, -1), (4, 0)\}$ is not a function: the input 3 is paired with both 5 and -1 . Deleting either pair makes it a function. The circle $x^2 + y^2 = 1$ is not a function either; the input $x = 0$ yields both $y = 1$ and $y = -1$.

Key idea. A function assigns to each input exactly one output. The domain is the set of inputs for which the assignment is defined; for each such input the output is determined.

When a function is given by a rule, $f(x)$ denotes the output at input x . The symbol $f(2)$ means the output of f at input 2, a single number. It does not mean f multiplied by 2.

Example 1.2.1. (*evaluating a function*) Let $f(x) = x^2 - 3x + 1$. Then

$$f(2) = 4 - 6 + 1 = -1, \quad f(0) = 1, \quad f(-1) = 1 + 3 + 1 = 5.$$

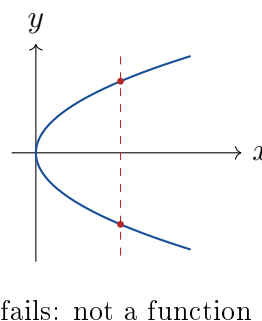
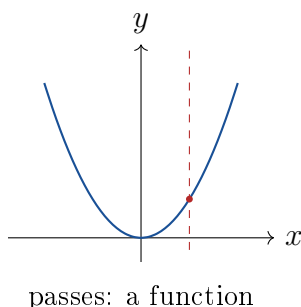
And $f(a + 1) = (a + 1)^2 - 3(a + 1) + 1 = a^2 - a - 1$ after expanding. Evaluating at an expression means substituting the entire expression for x .

Common pitfall. $f(a + b)$ is almost never $f(a) + f(b)$. For $f(x) = x^2$, $f(1 + 2) = f(3) = 9$, while $f(1) + f(2) = 1 + 4 = 5$. The function acts on the entire input at once, not on its parts separately.

1.2.1 The Vertical Line Test

The function property has a graphical reading. “Each input gives one output” means that above any point on the x -axis the graph has at most one point. A vertical line is the set of points sharing one x -value, which gives the test directly.

Key idea. *Vertical line test.* A graph represents a function if and only if no vertical line crosses it more than once.



Example 1.2.2. (*applying the test*) The graph of $y = |x|$ is a V opening upward; every vertical line hits it once, so it is a function. The graph of $x = y^2$ is a sideways parabola; the vertical line $x = 1$ hits it at $(1, 1)$ and $(1, -1)$, so it is not a function.

1.3 Fitting a Function to a Cloud

The two ideas now meet. Real data forms a point cloud with noise; it is almost never an exact function. A function can nonetheless be laid through the cloud as a *model*: a predictable rule that follows the trend closely enough to forecast, summarize, and reason about the data.

Key idea. A point cloud is the data. A function fit through it is a *model*. The model is not the data; it is the predictable rule chosen to represent the data so that one can compute and forecast with it.

The first decision is the *shape*. A cloud rising in a straight band calls for a linear model. A cloud curving upward like a bowl suggests a parabola. A cloud that climbs quickly and then levels off suggests a square-root or logarithmic shape. The catalog of basic shapes is the library of *parent functions*, and each can be reshaped to fit by the four operations that follow.

Interactive. A hidden transformed parent function generates a noisy point cloud. Identify the parent-function family, then tune $y = A f(b(x - h)) + k$ until the curve threads the cloud. The fit score reports the root-mean-squared error between curve and data; the hidden answer is deliberately approximate because the cloud carries noise.

https://bobovski66.github.io/Precalculus/demos/function_fit.html

Example 1.3.1. (*choosing a shape*) A cloud trends upward and bends, steep at first and then flatter. A straight line undershoots the early rise and overshoots the later leveling. The square-root shape $f(x) = \sqrt{x}$ has exactly this profile, fast and then flattening, so $\sqrt{x}$ is the parent to start from; the fitting work is sliding and stretching it onto the cloud.

1.4 Transformations: Bending One Curve into Another

Every function in the library can be reshaped by four operations, all captured in a single expression. Given a parent function f , form

$$g(x) = A f(b(x - h)) + k.$$

Each constant controls one geometric action. Fitting reduces to reading a cloud and setting these four numbers.

Key idea. The four constants of $g(x) = A f(b(x - h)) + k$.

- k : vertical shift. Adds to the output, moving the graph up ($k > 0$) or down ($k < 0$).
- h : horizontal shift. Moves the graph right ($h > 0$) or left ($h < 0$).
- A : vertical stretch by factor $|A|$; if $A < 0$, also a flip across the x -axis.
- b : horizontal stretch by factor $1/|b|$; if $b < 0$, also a flip across the y -axis.

1.4.1 Outside versus inside

The reliable bookkeeping is the outside/inside rule. Constants applied *outside* f (A and k) act on the output, so they move the graph *vertically* and behave as expected: $+k$ raises it, A scales its height. Constants applied *inside* f (b and h) act on the input, so they move

the graph *horizontally* and behave in reverse: $x - h$ shifts *right* by h , and multiplying the input by b *compresses* by b .

The picture. Outside changes act on the picture after it is drawn: raise it, scale its height. Inside changes act on the input before f receives it, so they warp the horizontal axis, and warping the input runs opposite to intuition. The parent produces its value at input 0; the transformed graph reaches that value where $b(x - h) = 0$, that is, at $x = h$. This is why $x - h$ moves the graph right rather than left.

Common pitfall. The horizontal shift is $x - h$, so $g(x) = f(x - 3)$ moves the graph *right* by 3, not left. The minus sign is built into the form. Read $f(x + 3)$ as $f(x - (-3))$, so $h = -3$ and the shift is left by 3.

Example 1.4.1. (*a single vertical shift*) With parent $f(x) = x^2$, the function $g(x) = x^2 + 3$ is the parabola lifted up 3 units. The vertex moves from $(0, 0)$ to $(0, 3)$; every output gains 3.

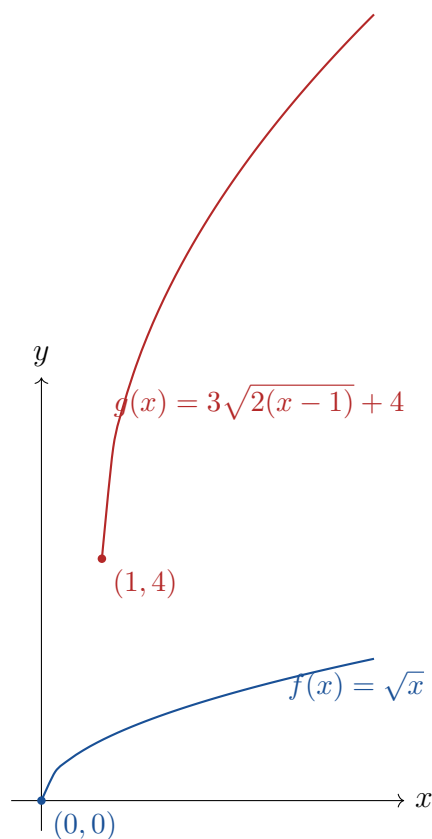
Example 1.4.2. (*a single horizontal shift*) With $f(x) = x^2$, the function $g(x) = (x - 2)^2$ slides the parabola right 2 units. The vertex moves from $(0, 0)$ to $(2, 0)$. Check: $g(2) = 0$, the old value at the old vertex input 0.

Example 1.4.3. (*stretch and reflect together*) With $f(x) = x^2$, the function $g(x) = -2x^2$ has $A = -2$: stretch vertically by 2 (twice as tall) and flip across the x -axis (opens downward). The point $(1, 1)$ on the parent becomes $(1, -2)$.

Example 1.4.4. (*all four at once*) Let $f(x) = \sqrt{x}$ and

$$g(x) = 3\sqrt{2(x - 1)} + 4.$$

Read off $A = 3$, $b = 2$, $h = 1$, $k = 4$. Starting from $\sqrt{x}$: compress horizontally by $\frac{1}{2}$, stretch vertically by 3, shift right 1, shift up 4. The parent begins at $(0, 0)$; the transformed curve begins at $(h, k) = (1, 4)$, since the inside is zero at $x = 1$ and the $+k$ raises the value to 4.



1.4.2 Order of operations on the graph

When several transformations combine, the correct order mirrors the evaluation of $g(x)$ at a number: work from the input outward. Apply the inside first (horizontal: the b compression, then the h shift), then the outside (vertical: the A stretch or reflection, then the k shift). Following one landmark point of the parent through each step keeps the order honest.

Example 1.4.5. (*tracking a point*) Take $f(x) = \sqrt{x}$ and the landmark $(4, 2)$ on it. Apply $g(x) = 3\sqrt{2(x-1)} + 4$ to that point's coordinates:

$$x: \quad 4 \xrightarrow{\div b} 2 \xrightarrow{+h} 3, \quad y: \quad 2 \xrightarrow{\times A} 6 \xrightarrow{+k} 10.$$

So $(4, 2)$ lands at $(3, 10)$. Confirm directly: $g(3) = 3\sqrt{2 \cdot 2} + 4 = 3 \cdot 2 + 4 = 10$. The two routes agree.

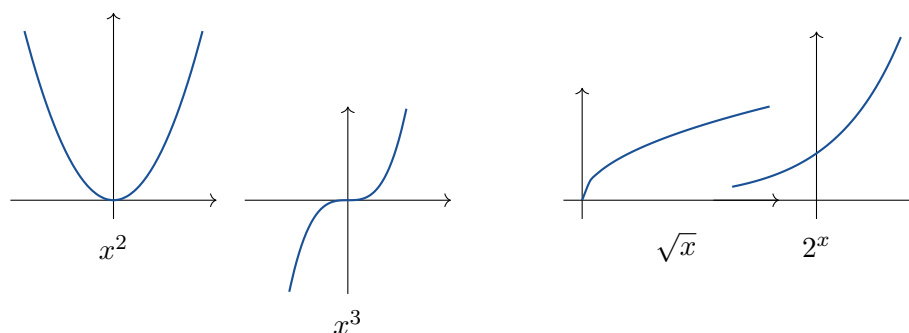
Common pitfall. The input coordinate transforms by the *inverse* operations in *reverse* order: to locate where an x lands, undo the inside. Dividing by b before adding h matches the algebra of $b(x - h)$; reversing those two steps misplaces the

point.

1.5 The Library of Parent Functions

Fitting is limited by the shapes one knows. The following parents recur throughout the course; their graphs should be memorized well enough to recognize in a cloud.

Name	Rule	Shape in a sentence
Constant	$f(x) = c$	flat horizontal line
Linear	$f(x) = x$	straight line through the origin, slope 1
Absolute value	$f(x) = x $	V opening up, corner at origin
Quadratic	$f(x) = x^2$	bowl opening up, vertex at origin
Cubic	$f(x) = x^3$	S-curve, flat at origin
Square root	$f(x) = \sqrt{x}$	half-parabola, fast then flattening
Reciprocal	$f(x) = 1/x$	two branches, axes as asymptotes
Exponential	$f(x) = 2^x$	flat then explosive growth



Example 1.5.1. (*matching a cloud to a parent*) A cloud lies entirely in the first quadrant, rising steeply near zero and then bending to a gentle climb. Among the parents, only $\sqrt{x}$ has this steep-then-gentle profile beginning at the origin. Start from $\sqrt{x}$ and fit A, b, h, k .

1.6 Exercises

Exercises 1.6

1.6.1. For the relation $\{(-1, 3), (0, 0), (2, 3), (2, -3), (5, 1)\}$, state the domain and

range, and decide whether it is a function. Justify.

1.6.2. Decide whether each graph could pass the vertical line test, and say what the test is checking: (a) a circle; (b) a downward parabola; (c) the letter “S” lying on its back; (d) a single steep upward line.

1.6.3. Let $f(x) = 2x^2 - x + 3$. Compute $f(0)$, $f(-2)$, and $f(t+1)$, simplifying the last completely.

1.6.4. Give a relation that is *not* a function, then change exactly one ordered pair to make it a function. Explain why your change works.

1.6.5. Starting from $f(x) = x^2$, write the rule for the function obtained by shifting left 4, stretching vertically by 3, and shifting down 1. Then state the vertex.

1.6.6. Describe in words every transformation taking $f(x) = |x|$ to $g(x) = -\frac{1}{2}|x+2|+5$, in the correct order, and give the coordinates of the corner of g .

1.6.7. The point $(9, 3)$ lies on $f(x) = \sqrt{x}$. Find where it lands under $g(x) = 2\sqrt{x-1}+4$ by tracking the coordinate through each step, then confirm by direct substitution.

1.6.8. Explain, using the outside/inside rule, why $f(x-3)$ shifts the graph right while $f(x)+3$ shifts it up. Why do the two “+3” moves go in such different directions?

1.6.9. A point cloud rises in a nearly straight band but with clear scatter. Explain the difference between the cloud and a linear function fit through it, and state what the word *model* means in this context.

1.6.10. For $g(x) = Af(b(x-h))+k$, suppose $A < 0$ and $b < 0$. Describe both reflections that occur and name the two axes involved.

1.6.11. A cloud opens upward like a bowl with its lowest point near $(2, -3)$. Propose a parent function and initial values of A, b, h, k to start a fit, and explain each choice.

1.6.12. Open the Function Fit Lab. Generate a cloud, identify the parent family, and record the four constants you settled on with the resulting fit score. State in one sentence which constant was hardest to determine and why.